

are combined to meet the needs and desires of each animal species. Enclosures are built to provide a comfortable and healthy home. The system provides livestock with the freedom to move freely in the virtual world, providing them with better treatment than they would be in the real world. Each enclosure is filled with independently filtered air to keep communicable diseases and parasites from spreading throughout the facility.

Biosensors are being used for animal health management. New wearable technologies, such as tracking collars and electronic saddles, are being used in animals, pets and livestock for their healthier upbringing. With the help of IoT-based movement sensors and microphones, farm owners can check the well-being and location of their livestock through smartphones.

Agricultural drones

Drones with such sensors as thermal, camera, lidar and multispectral are bringing major benefits to the agri-

cultural industry. Agricultural drones with sensors allow farmers to see their fields from the sky. Information and data collected from the drones reveal such issues as irrigation problems, soil variation, and pest and fungal infestations. The difference between healthy and unhealthy plants is sometimes difficult to detect with naked eyes. Multispectral images can assist in differentiating between healthy and unhealthy plants, and enable farmers to take timely actions.

Many companies are manufacturing drones for agricultural applications. For example, advanced autopilot drone from Drotek features advanced processors and latest sensor arrays working with a real-time operating system that can deliver incredible performance.

DJI manufactures a variety of high-tech drones. Agras MG-1S, also called the wonder drone, from DJI is a highly-advanced drone spraying system that has GPS, flow sensors, triple radar sensors and fully-programmable flight paths. Agras octocopter is designed



Fig. 5: Agras MG-1S by DJI

for precision variable rate application of liquid pesticides, fertilisers and herbicides with good efficiency and manageability.

Multispectral imaging system. Multispectral sensors collect visible (red, green and blue) wavelengths as well as infrared (IR) radiation and ultraviolet (UV) light. These are used to monitor plant health, identify deficiencies and pest damage, optimise fertilisation and assess water quality.

Sentera manufactures sensors to capture precise, radiometrically-accurate plant-level vegetative indices. The double 4K sensor from Sentera

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